

Anuj Patil

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Summary

Backend engineer (B.Tech Information Technology, 2026) specialising in Java/Python microservices and AI-integrated systems. Currently building Spring Boot microservices in production at Codologs. Shipped CodeEcoScan (live public API for Python energy profiling), Agent Sentinel (multi-stage misinformation pipeline, Mumbai Hacks 2025), and a pgvector + Redis RAG document engine — owning architecture, deployment, and observability end-to-end across each project.

Experience

Software Development Intern (Remote) Jan 2026 – May 2026

[Codologs] — Navi Mumbai / Remote

- Engineered RESTful API modules for core business features using Spring Boot 3 — including request routing, input validation, and service-layer logic — shipped to Codologs' production backend and consumed by downstream services within the same sprint cycle.
- Designed and delivered an authentication and session management microservice with JWT-based token lifecycle (issuance, validation, and revocation), deployed as a standalone service consumed by other backend modules.
- Introduced Redis response caching on high-frequency read endpoints, cutting p95 latency from ~320ms to ~95ms on cached routes by eliminating redundant database round-trips for repeated queries.
- Delivered 3 backend microservices now running in Codologs' production system — covering auth/session handling, a core business logic module, and a supporting service layer — each independently deployable via Spring Boot, with error logging monitored by the team in real-time.

Education

Bachelor of Technology — Information Technology 2022 – 2026

Pillai College of Engineering, University of Mumbai · CGPA: 7.7 / 10.0

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C

Backend: Spring Boot, FastAPI, Node.js, REST API Design

Frontend: React.js, HTML/CSS

Databases: PostgreSQL (pgvector), MySQL, Redis

AI / ML: LangChain, RAG, LoRA Fine-Tuning, ChromaDB

DevOps / Cloud: Docker, Docker Compose, GitHub Actions, AWS (S3, EC2 basics), Railway, Render

Tools: Git, IntelliJ IDEA, VS Code, Jupyter Notebook, Postman, Sentry

Core CS: Data Structures & Algorithms, OOP, System Design fundamentals, Multithreading

Projects

CodeEcoScan — Lead Developer · [Live public API](#)

- Engineered a static analysis tool that estimates energy risk and carbon footprint of Python workloads, scoring code across loop complexity, recursion depth, import weight, and I/O patterns using a custom AST-based engine.
- Architected a real-time carbon emissions estimator parameterised by hardware profile (Laptop / Cloud VM / GPU Server), runtime, and run frequency — converting energy draw to kgCO₂/day.
- Deployed a live public API on Render serving analysis results in under 200ms for typical Python files; built a terminal-inspired React frontend with syntax-highlighted heatmap overlays and per-category score breakdowns.

Tech: Python (FastAPI), React.js, AST analysis, Render

Agent Sentinel — Lead Backend Engineer · Presented at Mumbai Hacks 2025

- Built an AI classification pipeline that processes viral claims end-to-end in under 4 seconds — extracting claims, retrieving evidence via GDELt, scoring with RoBERTa, and auto-publishing high-confidence verdicts while routing low-confidence results to a human review queue.
- Architected the multi-stage backend pipeline — claim extraction, GDELt-backed fact retrieval, RoBERTa classification, and structured verdict generation with confidence scores — designed for B2G and B2B misinformation monitoring use cases.
- Selected to present at Mumbai Hacks 2025 (December) — one of Maharashtra's largest hackathons — competing among 200+ teams as a 2-member team.

Tech: Python, FastAPI, HuggingFace Transformers (RoBERTa), GDELt API

SmartDoc Query Engine — Full-Stack Developer / DevOps [LivePublicAPI](#)

- Architected a RAG system for secure PDF querying using Docker Compose with isolated service containers, enabling independent restarts and fault containment per service — separating the API, embedding, and database layers.
- Implemented vectorised document storage with PostgreSQL (pgvector) and Redis-based caching + rate limiting, reducing redundant embedding calls by ~65% for repeat queries on previously-indexed documents.
- Added full observability via Sentry with structured logging of model performance and edge-case parsing failures.

Tech: Python (FastAPI), React.js (Vite), PostgreSQL (pgvector), Redis, Docker Compose, Sentence Transformers

AI Tutor with Voice Cloning & Context-Managed Persona — Backend Developer (4-member team)

- Built a custom-trained AI tutor that answers course-specific questions while maintaining instructor tone and style through context-window-managed prompt injection and LoRA fine-tuning, evaluated on instructor-curated course question sets.
- Integrated voice cloning (XTTS) with RAG-based retrieval over ChromaDB, enabling the tutor to respond in the instructor's synthesised voice using LoRA fine-tuned models.

Tech: Python, FastAPI, Streamlit, RAG, LoRA Fine-Tuning, XTTS (Voice Cloning), ChromaDB, MLX

Changelog-AI [Live Public API](#)

- Developed an end-to-end NLP pipeline using Gemini models and LangChain to transform raw GitHub commit metadata into structured, user-centric release notes automatically.
- Deployed backend on Railway with automatic restart configuration and Sentry observability to monitor runtime performance and capture edge-case parsing failures in real-time.

Tech: Python, LangChain, Gemini API, Railway, Sentry

Hackathons & Achievements

Mumbai Hacks 2025 — Participant December 2025

Mumbai, Maharashtra

- Participant | December 2025 Built and presented Agent Sentinel, an autonomous misinformation detection system, as a 2-member team at one of Maharashtra's largest annual hackathons.

Smart India Hackathon (SIH) — Internal Round Participant 2025

Pillai College of Engineering, New Panvel

- Cleared the internal college selection round.

Hack4Innovation Hackathon — Selected Participant March 2026

Vivekanand Education Society's Institute Of Technology (VESIT), Mumbai

Selected as a core member of Team Code Titans for a competitive hackathon hosted by Vivekanand Institute of Technology in collaboration with Rotary Club Mumbai — demonstrating analytical problem-solving and technical execution.